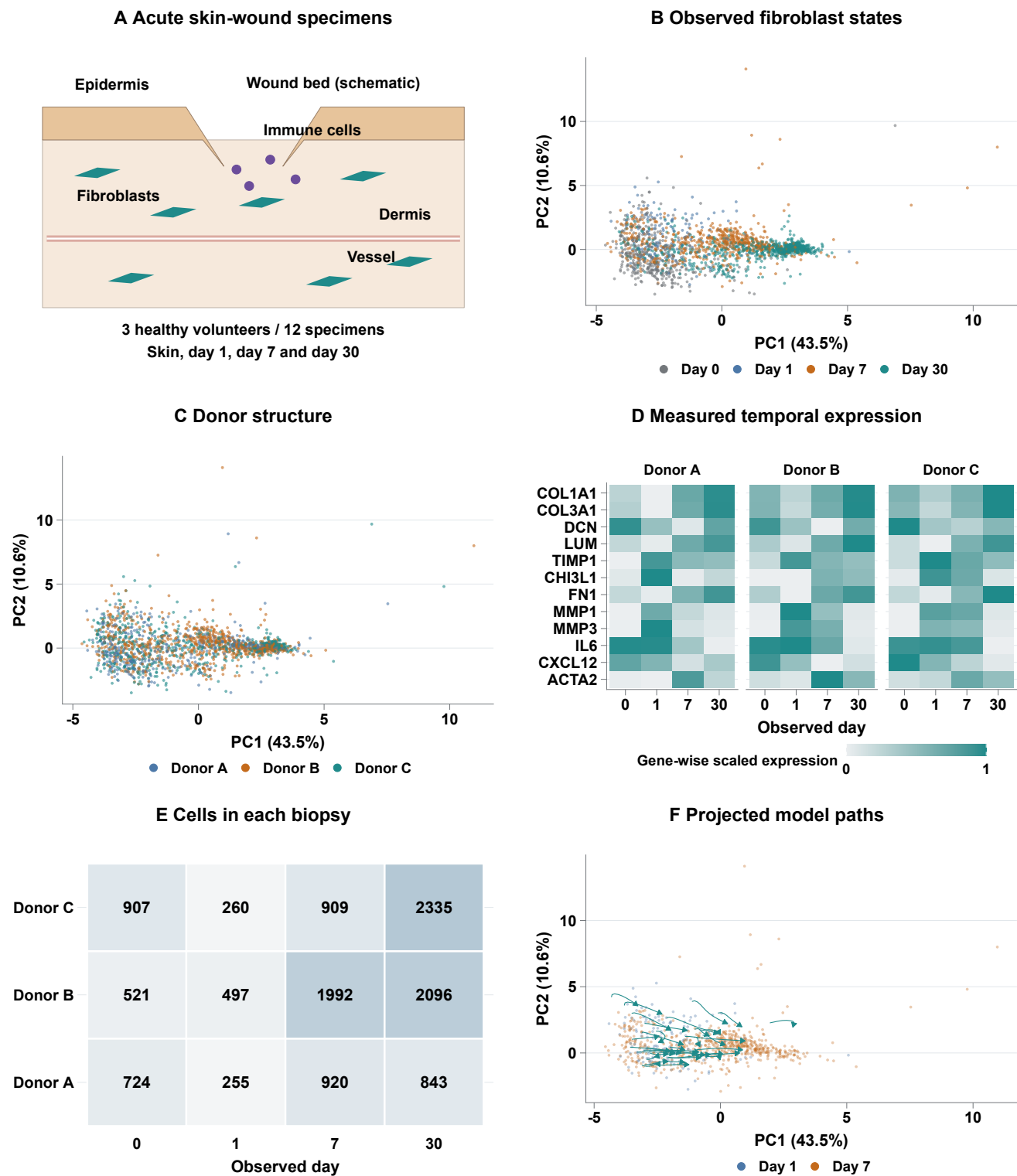
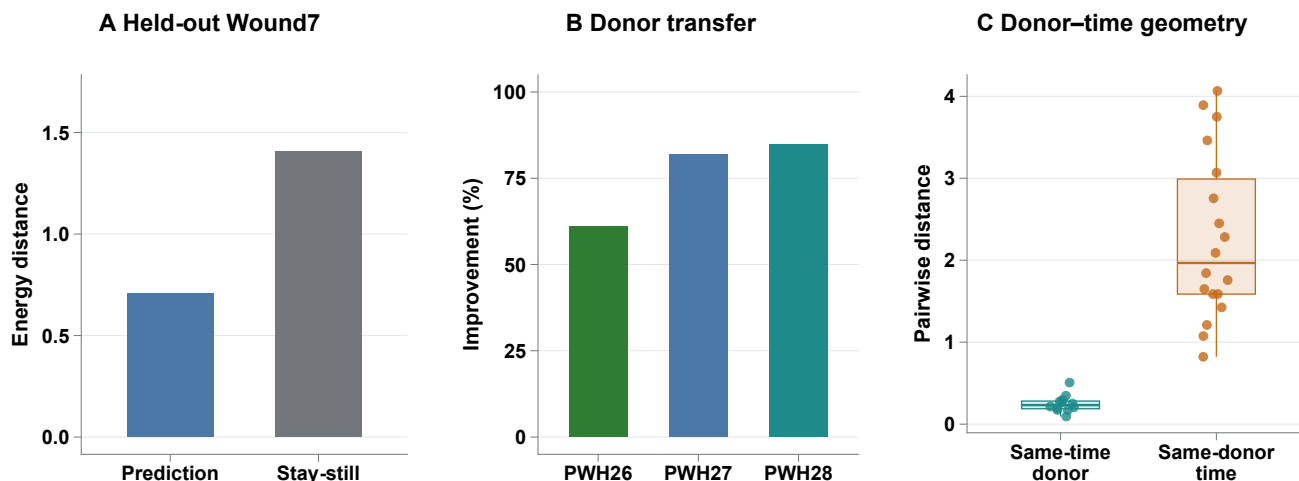


Figure 1



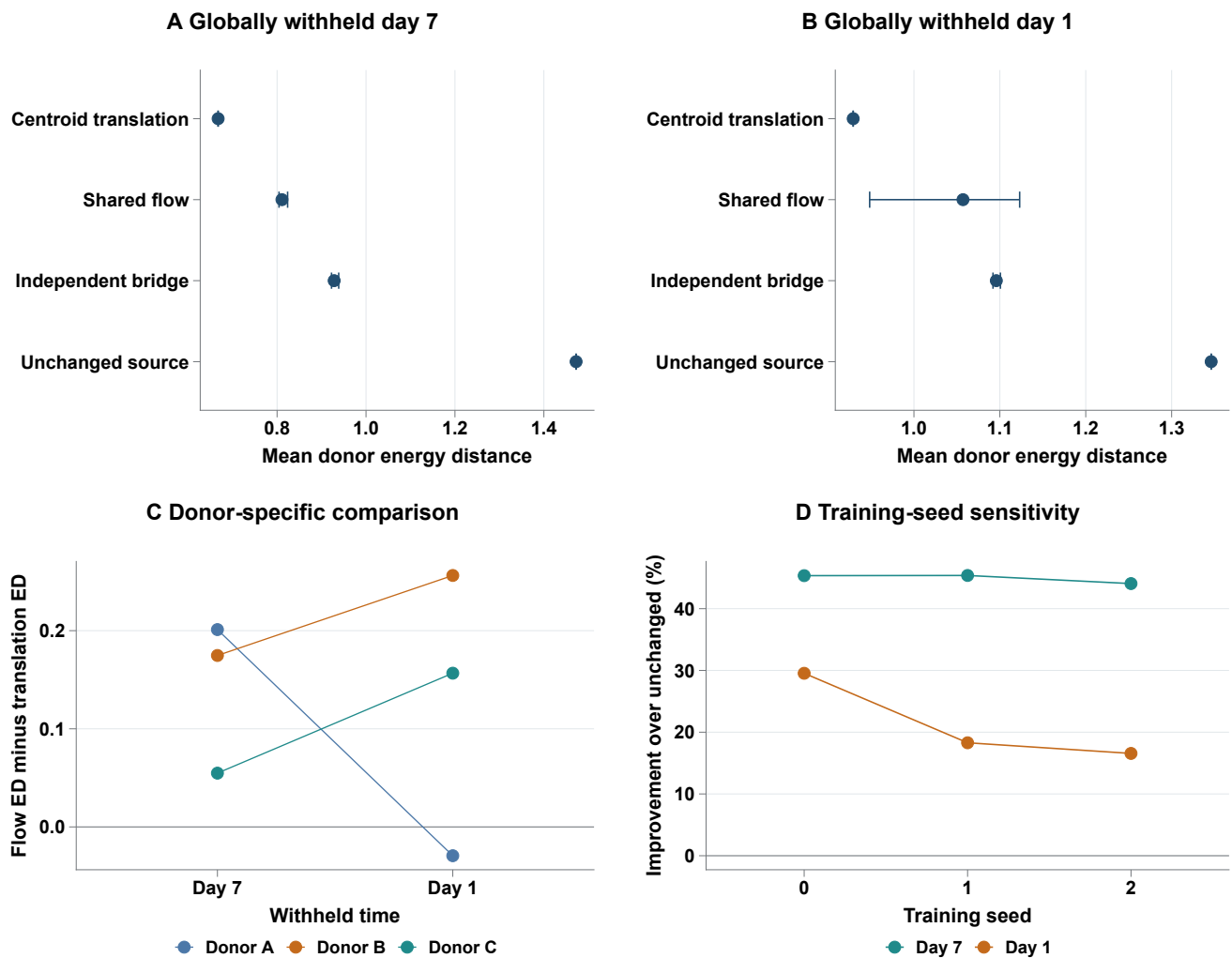
Observed acute-wound material and its relation to model coordinates. (A) Conceptual tissue anatomy and the four biopsy times in three healthy volunteers; no histology is inferred. (B) A uniform 2,000-cell subset of 12,259 observed fibroblasts, colored by time. PCA and standardization use only day-7 training cells; axes explain 43.5% and 10.6% of training variance. (C) The identical cells colored by donor. (D) Twelve measured genes in every donor–time biopsy, using log-transformed pseudobulk counts per 10,000 and a separate zero-to-one scale for each gene. Colors do not compare absolute expression across genes. (E) Retained fibroblast counts per biopsy; cells are not independent donors. (F) Thirty-six seed-0 model paths from observed day-1 cells to the day-7 rank coordinate, over the observed day-1/day-7 display. Arrows indicate numerical motion, not tracked cell fates. All errors use the full 15-dimensional coordinates.

Figure 2



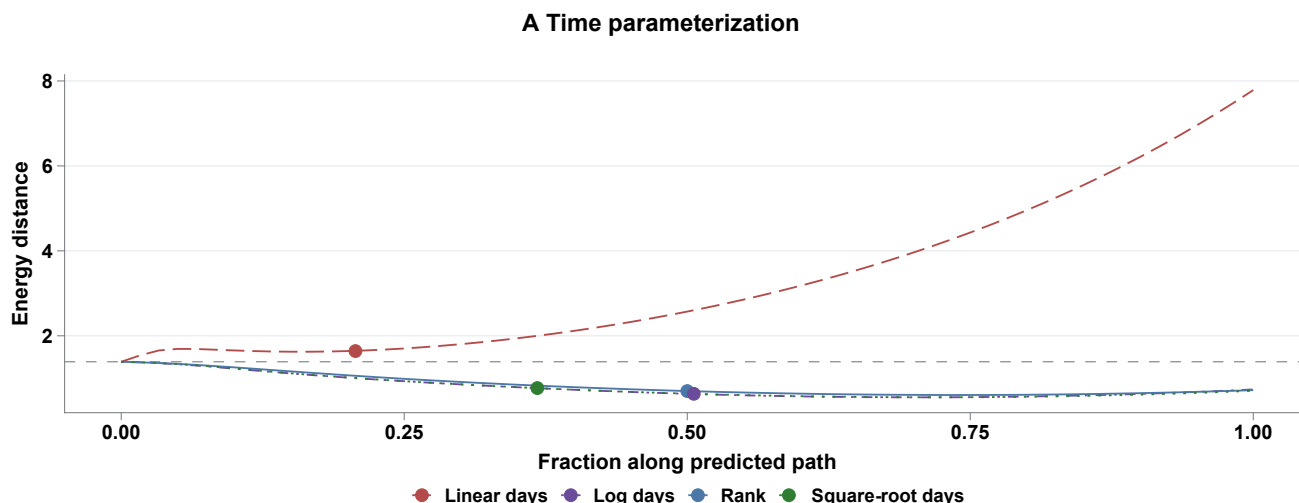
Global Wound7 reconstruction, donor transfer and donor-time geometry. (A) Bars show predicted and stay-still energy distances to pooled Wound7 after excluding that time globally; relative improvement is 100 times (stay-still minus predicted distance) divided by stay-still distance, giving 49.9%. (B) Each bar is a genuinely held-out donor, with all four times retained in the other two donors; the mean relative improvement is 75.9%. Bars are point estimates without confidence intervals. (C) Points show 12 between-donor same-time distances and 18 within-donor between-time distances. Boxes span the interquartile range, centre lines are medians, and whiskers extend to extreme points within 1.5 interquartile ranges. The temporal/donor median ratio is 8.46 (10,000-draw descriptive pair-distance bootstrap 95% interval 6.02–13.11). All distances use training-standardized latent coordinates. Shared donors make these pairs dependent; three healthy donors do not provide DFU prediction validation.

Figure 3



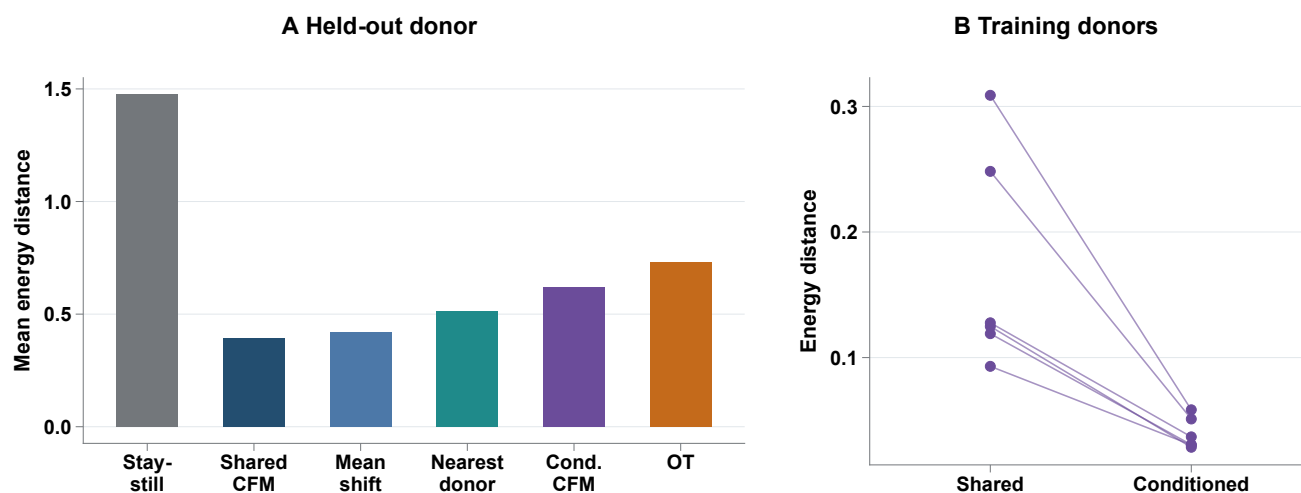
Simple baselines for globally missing times. All methods use the same frozen representation and training exclusions; error averages full-cell empirical energy distances within the three donors. (A) Withheld day 7, predicted from day 1 with day 30 as the observed anchor. (B) Withheld day 1, predicted from intact skin with day 7 as anchor. Points show the three-seed mean error; horizontal ranges show seed minima and maxima, not confidence intervals. Both tasks use the fixed rank-time interpolation fraction one half. (C) Donor-specific seed-averaged field error minus centroid-translation error; positive values favor translation. The later anchor belongs to the same donor, so this is retrospective interpolation rather than source-only donor transfer. (D) Each field seed's improvement over the corresponding unchanged source. Seeds and repeated distance evaluations provide no additional biological replication.

Figure 4



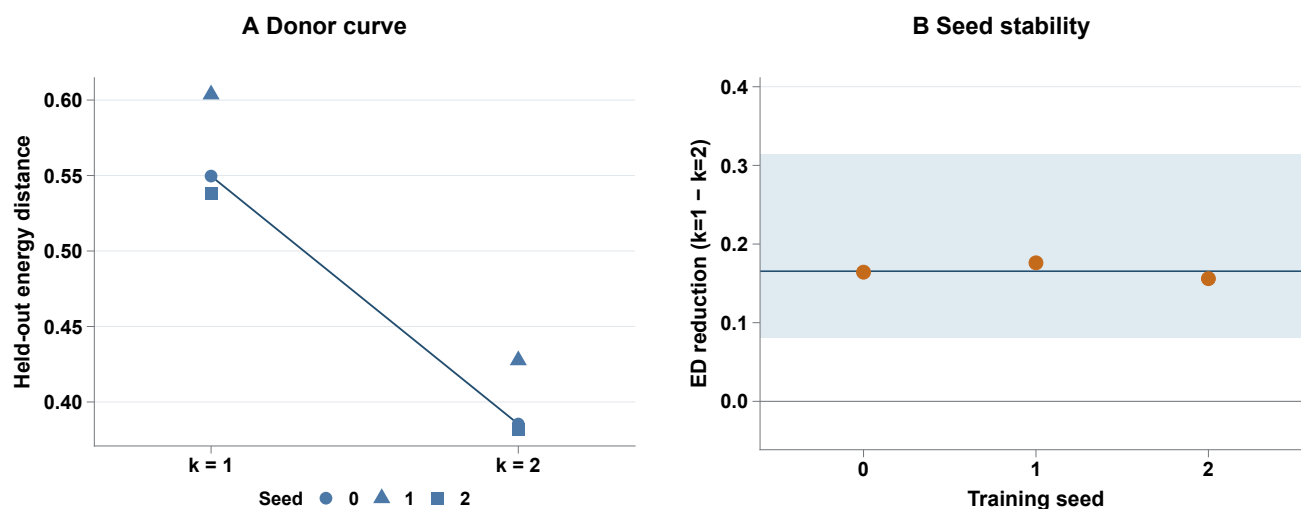
Time-coordinate sensitivity under global Wound7 holdout in three donors. (A) Curves show energy distance along the Wound1-to-Wound30 predicted segment for rank, linear-day, square-root-day and log-day coordinates. The horizontal axis runs from the Wound1 source (0) to the Wound30 coordinate (1); line styles and colours identify the transforms. Dots are placed at each nominal Wound7 fraction with the distance evaluated at the nearest of 61 stored grid positions from 60 RK4 steps. The grey dashed line is the stay-still distance, 1.3879. Linear days gives -18.2% relative improvement at the target; the other axes have gains above the single split-half reference, 0.0061. The full path is a retrospective diagnostic, and its minimum is not substituted for the Wound7 result. No tested axis diverges; curves do not represent confidence bands.

Figure 5



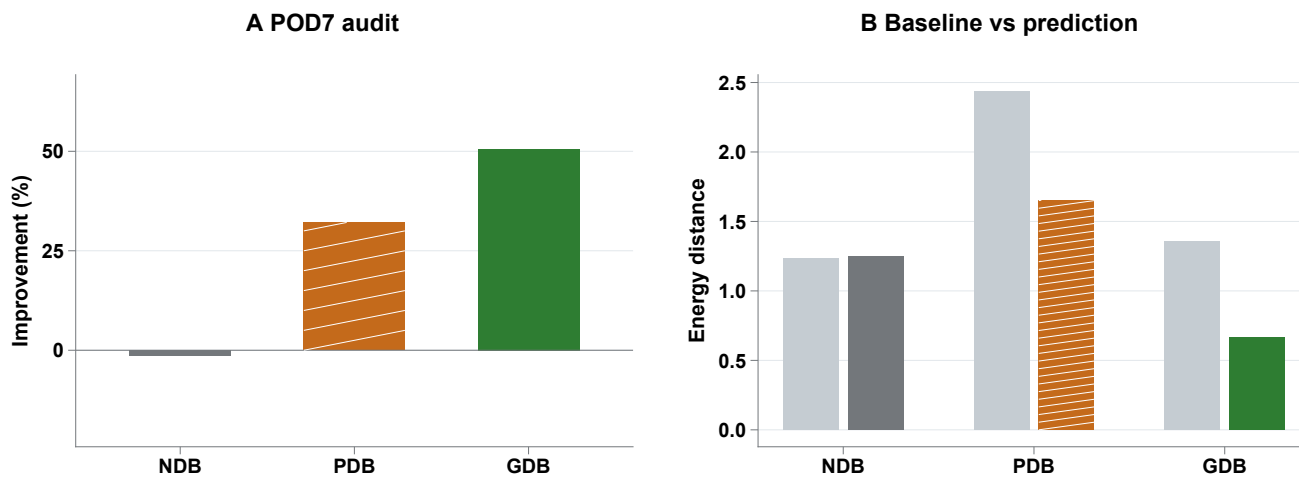
Method comparison across three held-out donors. (A) Bars are arithmetic mean energy distances for stay-still and five predictors evaluated on the same donor folds. CFM means conditional flow matching; OT is the entropic optimal-transport displacement extension. Shared CFM has the lowest point estimate (0.395). Each predictive method improves over stay-still by more than the single split-half target reference in 3/3 folds; this is a descriptive criterion, not a significance test or population method ranking. (B) Each line connects shared and donor-conditioned errors for one of six training-donor comparisons (two retained donors in each of three fits). Conditioning improves all six. These training-donor fits share donors and do not measure transfer to an unseen donor. Neither panel shows uncertainty intervals.

Figure 6



Donor-count comparison and training-seed variation. (A) k is the number of training donors. The line joins seed-0 mean held-out energy distances, 0.5496 at $k=1$ and 0.3854 at $k=2$; symbols identify seeds 0, 1 and 2. At $k=1$, the two single-training-donor choices are averaged within each held donor before equally weighting the three held donors. (B) Points are seed-specific reductions in mean energy distance (ED), $k=1$ minus $k=2$. The horizontal line is the three-seed mean, 0.165. Shading is a percentile 95% interval [0.081, 0.315] from 4,000 resamples of three held-donor blocks after seed averaging; it is not an interval for each seed point. Evaluation subsampling is fixed across seeds. The largest configuration-specific range is 0.368. Overlapping training sets limit interval interpretation, and two k values cannot identify a plateau or required cohort size.

Figure 7



Mouse POD7 computational audit for the source-labelled NDB, PDB and GDB arms. POD7 is excluded from fitting and scaling; rank-time fields trained at POD0, POD2 and POD30 predict from the observed POD2 source. (A) Relative energy-distance improvements over stay-still are -1.5% , $+32.1\%$ and $+50.6\%$. Hatching marks PDB, whose POD7 target fails the endpoint-distance ordering diagnostic. (B) Within each arm, the left grey bar is stay-still and the right coloured bar is prediction; hatching carries the same diagnostic meaning. The screen compares endpoint distances and does not prove or disprove geodesic interpolation. Each arm has one animal per time point and 4,182/7,002 mapped panel genes. Bars have no uncertainty intervals; cell counts do not supply independent animal replication or cross-species clinical validation.